Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Vander Meulen Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
20250 Avena RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0205-0110-0001-0000

B. OPERATORS

Vander Meulen, William

Operator name: Vander Meulen, William

Telephone no.: (209) 838-2606 Landline, (209) 988-7405 Cellular

Mailing Address Number and Street	City	State	Zip Code
20250 Avena RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

Vander Meulen, William

Legal owner name: Vander Meulen, William

Telephone no.: (209) 838-2606 Landline, (209) 988-7405 Cellular

Mailing Address Number and Street	City	State	Zip Code
20250 Avena RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	100	100	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	100	100	0	0
Average number	0	0	100	100	0	0
Avg live weight (lbs)	0	0	750	550		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1,860.76 *tons per reporting period*

Total nitrogen from manure: 18,980.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 13,286.00 *lbs per reporting period*

Total phosphorus from manure: 3,212.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag 1	Ground water
Ag 2	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/27/2023	Liquid commercial fertilizer *UN32*	8,400.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	22,431.36	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	22,431.36	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	40	40	2	manure	X205-X110-XX02-XXXX
2	10	10	1	none	X205-X110-XX02-XXXX
3	7	7	2	manure	X205-X110-XX02-XXXX
4	15	15	2	manure	X205-X110-XX02-XXXX
5	20	20	2	manure	X205-X140-XX02-XXXX
6	30	30	2	manure	X205-X140-XX02-XXXX
Totals for areas that were used for application	112	112	10		
Totals for areas that were not used for application	10	10	1		
Land application area totals	122	122	11		

B. CROPS AND HARVESTS

1

Field name: 1

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	550.00 *ton*	Dry-weight		62.3	9,300.00	236.00	18,200.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	13.75	96.42	2.45	188.69	1,078.22

1

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,000.00 *ton*	Dry-weight		72.0	16,500.00	2,660.00	20,800.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	25.00	231.00	37.24	291.20	1,344.00

2

Field name: 2

10/28/2009: Pasture

Crop: Pasture Acres planted: 10 Plant date: 10/28/2009

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	100.00 *ton*	Dry-weight		70.0	10,000.00	2,000.00	10,000.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	150.00	30.00	150.00	0.00
Total actual harvest content	10.00	60.00	12.00	60.00	0.00

3

Field name: 3

3

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 7 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	100.00 *ton*	Dry-weight		65.5	10,200.00	2,260.00	32,400.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.29	100.54	22.28	319.37	1,005.43

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 7 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	170.00 *ton*	Dry-weight		66.2	14,400.00	2,650.00	19,400.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.29	236.41	43.51	318.49	1,329.79

4

Field name: 4

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	200.00 *ton*	Dry-weight		71.2	11,200.00	2,310.00	20,900.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	13.33	86.02	17.74	160.51	714.24

4

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	370.00 *ton*	Dry-weight		64.4	13,500.00	2,210.00	19,200.00		8.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.67	237.10	38.81	337.20	1,440.14

5

Field name: 5

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	280.00 *ton*	Dry-weight		64.0	14,300.00	3,100.00	25,300.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.00	144.14	31.25	255.02	1,108.80

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	490.00 *ton*	Dry-weight		66.0	14,600.00	3,180.00	14,600.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.50	243.24	52.98	243.24	1,299.48

6

Field name: 6

10/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	420.00 *ton*	Dry-weight		62.4	11,300.00	2,600.00	21,800.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.00	118.97	27.37	229.51	1,084.38

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	740.00 *ton*	Dry-weight		63.4	13,600.00	2,970.00	13,800.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	24.67	245.56	53.63	249.17	1,336.14

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/26/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	102.18	86.36	231.40	0.00	150.00 *ton*
Application event totals		102.18	86.36	231.40	0.00	

1 - 06/05/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	11.88	0.00	0.00	567.79	5,040,000.00 *gal*
Application event totals		11.88	0.00	0.00	567.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	10.18	0.00	0.00	486.68	4,320,000.00 *gal*
Application event totals		10.18	0.00	0.00	486.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	200.00	0.00	0.00	0.00	
Application event totals		200.00	0.00	0.00	0.00	

1 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	10.18	0.00	0.00	486.68	4,320,000.00 *gal*
Application event totals		10.18	0.00	0.00	486.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	10.18	0.00	0.00	486.68	4,320,000.00 *gal*
Application event totals		10.18	0.00	0.00	486.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	10.18	0.00	0.00	486.68	4,320,000.00 *gal*
Application event totals		10.18	0.00	0.00	486.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	10.18	0.00	0.00	486.68	4,320,000.00 *gal*
Application event totals		10.18	0.00	0.00	486.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	10.18	0.00	0.00	486.68	4,320,000.00 *gal*
Application event totals		10.18	0.00	0.00	486.68	

2 - 10/28/2009: Pasture

Field name: 2

Crop: Pasture

Plant date: 10/28/2009

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 10/28/2009: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

2 - 10/28/2009: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	6.79	0.00	0.00	324.45	720,000.00 *gal*
Application event totals		6.79	0.00	0.00	324.45	

3 - 10/26/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	116.78	98.70	264.45	0.00	30.00 *ton*
Application event totals		116.78	98.70	264.45	0.00	

3 - 06/05/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	12.12	0.00	0.00	579.38	900,000.00 *gal*
Application event totals		12.12	0.00	0.00	579.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	9.70	0.00	0.00	463.51	720,000.00 *gal*
Application event totals		9.70	0.00	0.00	463.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	200.00	0.00	0.00	0.00	
Application event totals		200.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	9.70	0.00	0.00	463.51	720,000.00 *gal*
Application event totals		9.70	0.00	0.00	463.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	9.70	0.00	0.00	463.51	720,000.00 *gal*
Application event totals		9.70	0.00	0.00	463.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	9.70	0.00	0.00	463.51	720,000.00 *gal*
Application event totals		9.70	0.00	0.00	463.51	

3 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	9.70	0.00	0.00	463.51	720,000.00 *gal*
Application event totals		9.70	0.00	0.00	463.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	9.70	0.00	0.00	463.51	720,000.00 *gal*
Application event totals		9.70	0.00	0.00	463.51	

4 - 10/26/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	109.00	92.12	246.82	0.00	60.00 *ton*
Application event totals		109.00	92.12	246.82	0.00	

4 - 06/05/2023: Corn, silage

Field name: 4

Crop: Corn, silage Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	13.58	0.00	0.00	648.91	2,160,000.00 *gal*
Application event totals		13.58	0.00	0.00	648.91	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	11.32	0.00	0.00	540.76	1,800,000.00 *gal*
Application event totals		11.32	0.00	0.00	540.76	

07/07/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	200.00	0.00	0.00	0.00	
Application event totals		200.00	0.00	0.00	0.00	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	11.32	0.00	0.00	540.76	1,800,000.00 *gal*
Application event totals		11.32	0.00	0.00	540.76	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	11.32	0.00	0.00	540.76	1,800,000.00 *gal*
Application event totals		11.32	0.00	0.00	540.76	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	11.32	0.00	0.00	540.76	1,800,000.00 *gal*
Application event totals		11.32	0.00	0.00	540.76	

4 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	11.32	0.00	0.00	540.76	1,800,000.00 *gal*
Application event totals		11.32	0.00	0.00	540.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 1	Ground water	11.32	0.00	0.00	540.76	1,800,000.00 *gal*
Application event totals		11.32	0.00	0.00	540.76	

5 - 10/26/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	95.37	80.60	215.97	0.00	70.00 *ton*
Application event totals		95.37	80.60	215.97	0.00	

5 - 06/05/2023: Corn, silage

Field name: 5

Crop: Corn, silage Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	20.07	0.00	0.00	0.00	2,880,000.00 *gal*
Application event totals		20.07	0.00	0.00	0.00	

06/30/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	200.00	0.00	0.00	0.00	
Application event totals		200.00	0.00	0.00	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	16.05	0.00	0.00	0.00	2,304,000.00 *gal*
Application event totals		16.05	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	16.05	0.00	0.00	0.00	2,304,000.00 *gal*
Application event totals		16.05	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	16.05	0.00	0.00	0.00	2,304,000.00 *gal*
Application event totals		16.05	0.00	0.00	0.00	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	16.05	0.00	0.00	0.00	2,304,000.00 *gal*
Application event totals		16.05	0.00	0.00	0.00	

5 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	16.05	0.00	0.00	0.00	2,304,000.00 *gal*
Application event totals		16.05	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	16.05	0.00	0.00	0.00	2,304,000.00 *gal*
Application event totals		16.05	0.00	0.00	0.00	

6 - 10/26/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	90.83	76.77	205.69	0.00	100.00 *ton*
Application event totals		90.83	76.77	205.69	0.00	

6 - 06/05/2023: Corn, silage

Field name: 6

Crop: Corn, silage Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	16.05	0.00	0.00	0.00	3,456,000.00 *gal*
Application event totals		16.05	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	14.27	0.00	0.00	0.00	3,072,000.00 *gal*
Application event totals		14.27	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	200.00	0.00	0.00	0.00	
Application event totals		200.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	14.27	0.00	0.00	0.00	3,072,000.00 *gal*
Application event totals		14.27	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	14.27	0.00	0.00	0.00	3,072,000.00 *gal*
Application event totals		14.27	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	14.27	0.00	0.00	0.00	3,072,000.00 *gal*
Application event totals		14.27	0.00	0.00	0.00	

6 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	14.27	0.00	0.00	0.00	3,072,000.00 *gal*
Application event totals		14.27	0.00	0.00	0.00	

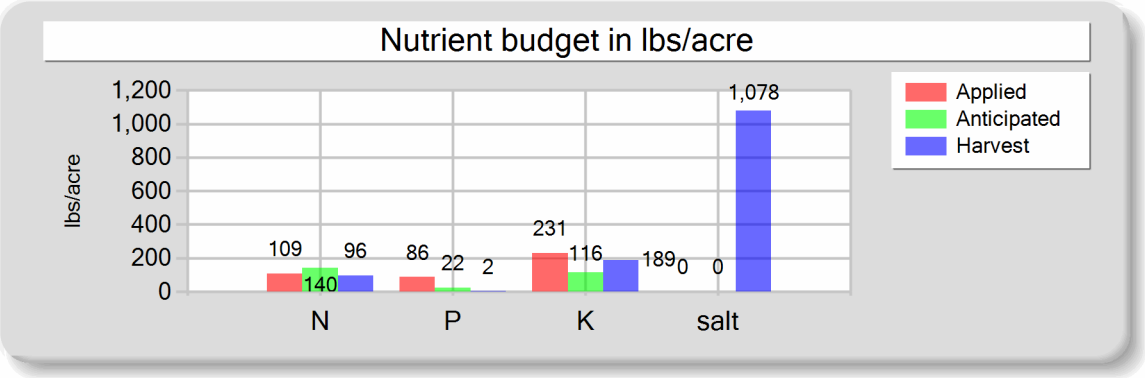
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag 2	Ground water	14.27	0.00	0.00	0.00	3,072,000.00 *gal*
Application event totals		14.27	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 10/26/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	102.18	86.36	231.40	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	109.18	86.36	231.40	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	96.42	2.45	188.69	1,078.22
Nutrient balance	12.77	83.92	42.71	-1,078.22
Applied to removed ratio	1.13	35.30	1.23	0.00

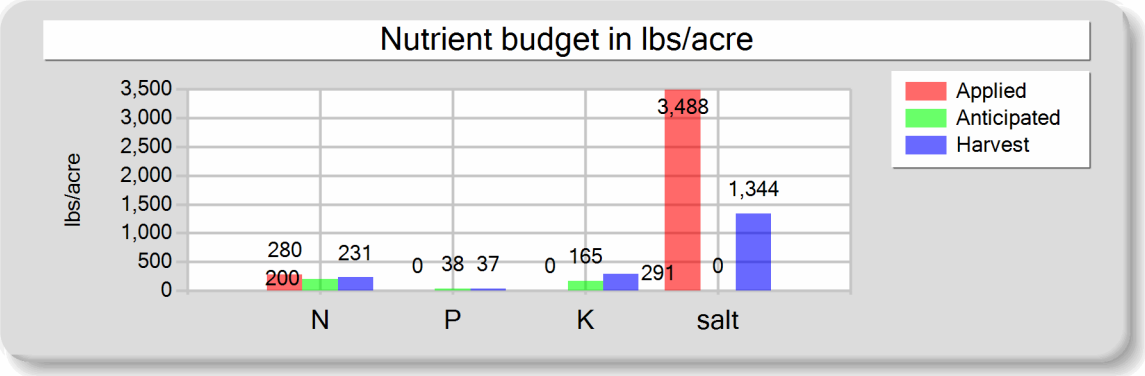
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

1 - 06/05/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	200.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	72.99	0.00	0.00	3,487.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	279.99	0.00	0.00	3,487.88
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	231.00	37.24	291.20	1,344.00
Nutrient balance	48.99	-37.24	-291.20	2,143.88
Applied to removed ratio	1.21	0.00	0.00	2.60

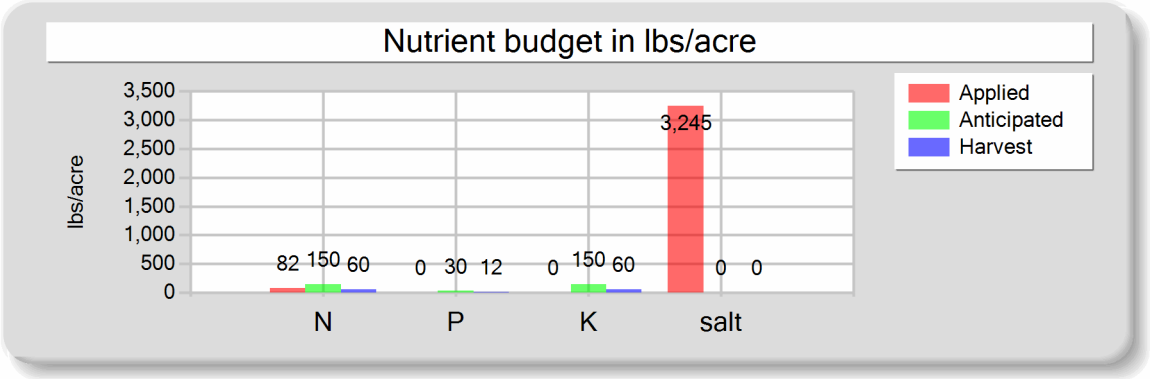
Fresh water applied	
30,960,000.00	*gallons*
1,140.15	*acre-inches*
28.50	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 10/28/2009: Pasture

Field name: 2 Crop: Pasture Plant date: 10/28/2009



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	67.89	0.00	0.00	3,244.54
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	81.89	0.00	0.00	3,244.54
Anticipated crop nutrient removal	150.00	30.00	150.00	0.00
Actual crop nutrient removal	60.00	12.00	60.00	0.00
Nutrient balance	21.89	-12.00	-60.00	3,244.54
Applied to removed ratio	1.36	0.00	0.00	0.00

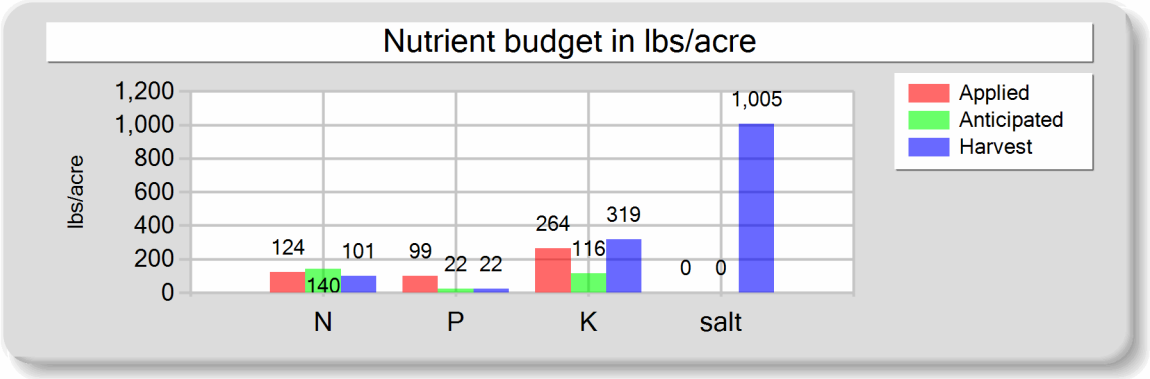
Fresh water applied	
7,200,000.00	*gallons*
265.15	*acre-inches*
26.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/26/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	116.78	98.70	264.45	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	123.78	98.70	264.45	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	100.54	22.28	319.37	1,005.43
Nutrient balance	23.24	76.42	-54.92	-1,005.43
Applied to removed ratio	1.23	4.43	0.83	0.00

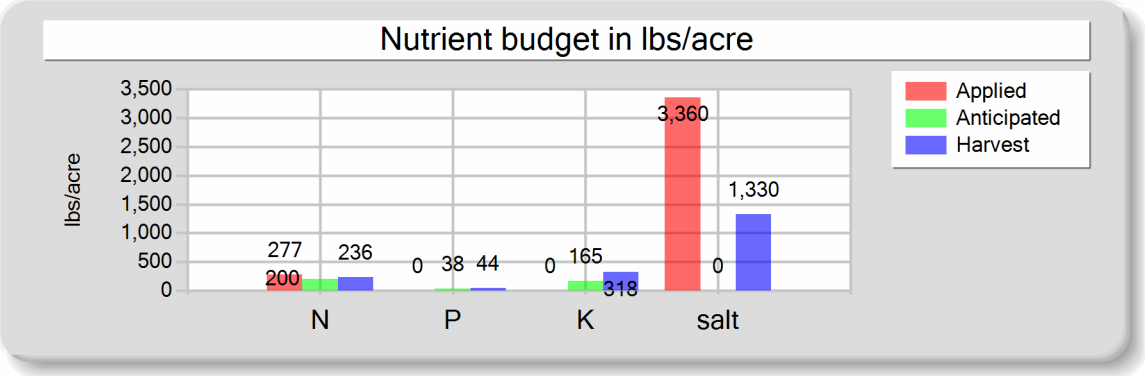
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

3 - 06/05/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	200.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	70.32	0.00	0.00	3,360.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	277.32	0.00	0.00	3,360.41
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	236.41	43.51	318.49	1,329.79
Nutrient balance	40.91	-43.51	-318.49	2,030.62
Applied to removed ratio	1.17	0.00	0.00	2.53

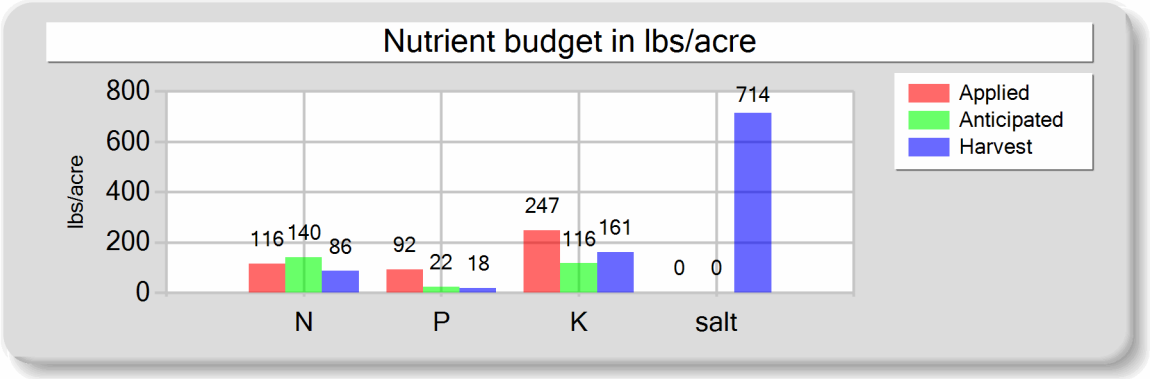
Fresh water applied	
5,220,000.00	*gallons*
192.23	*acre-inches*
27.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/26/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	109.00	92.12	246.82	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	116.00	92.12	246.82	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	86.02	17.74	160.51	714.24
Nutrient balance	29.98	74.38	86.31	-714.24
Applied to removed ratio	1.35	5.19	1.54	0.00

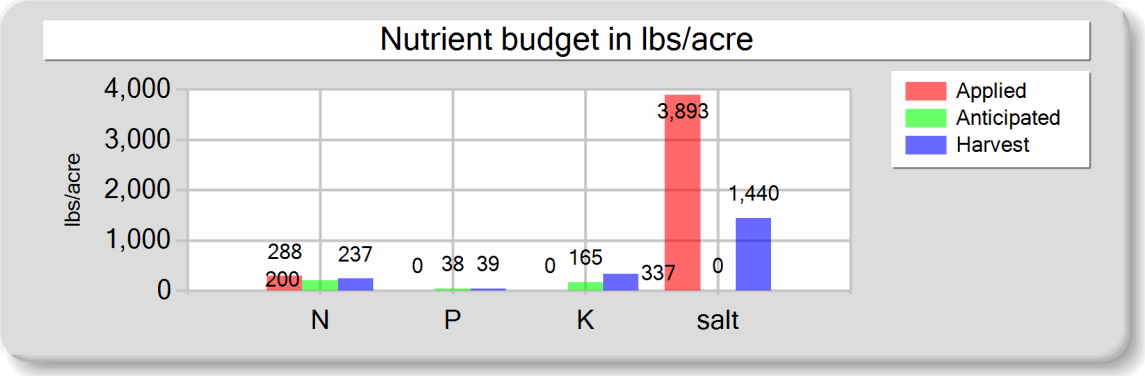
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

4 - 06/05/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	200.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	81.47	0.00	0.00	3,893.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.47	0.00	0.00	3,893.44
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	237.10	38.81	337.20	1,440.14
Nutrient balance	51.38	-38.81	-337.20	2,453.30
Applied to removed ratio	1.22	0.00	0.00	2.70

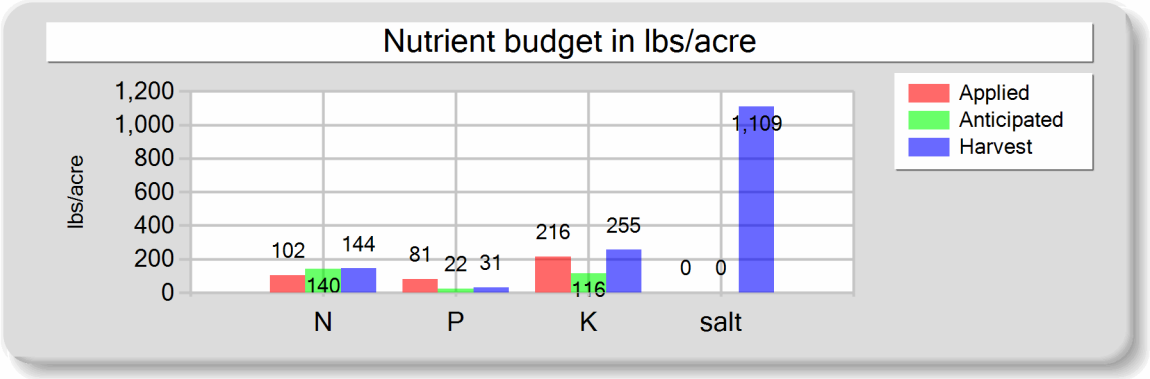
Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
31.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/26/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	95.37	80.60	215.97	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	102.37	80.60	215.97	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	144.14	31.25	255.02	1,108.80
Nutrient balance	-41.77	49.36	-39.05	-1,108.80
Applied to removed ratio	0.71	2.58	0.85	0.00

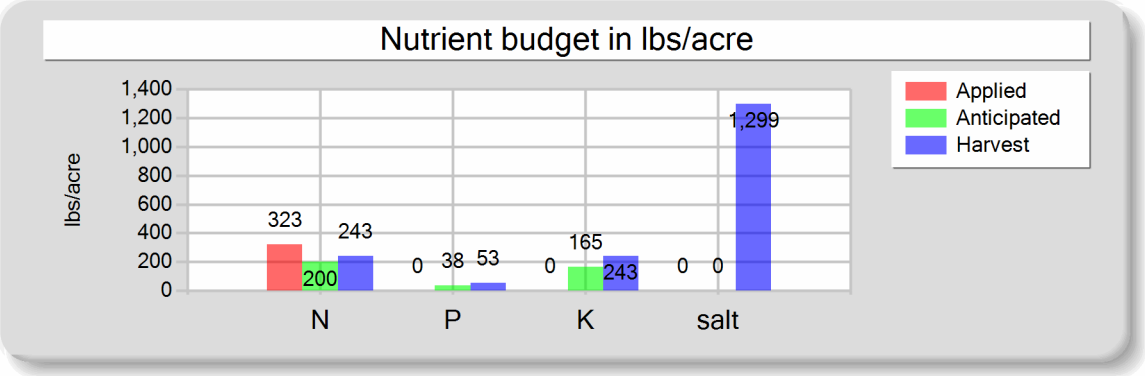
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

5 - 06/05/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	200.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	116.39	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.39	0.00	0.00	0.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	243.24	52.98	243.24	1,299.48
Nutrient balance	80.16	-52.98	-243.24	-1,299.48
Applied to removed ratio	1.33	0.00	0.00	0.00

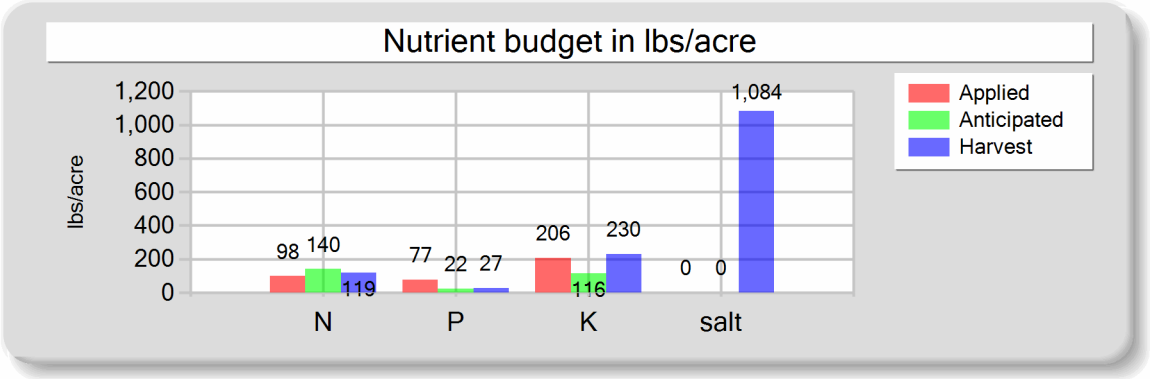
Fresh water applied	
16,704,000.00	*gallons*
615.15	*acre-inches*
30.76	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 10/26/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	90.83	76.77	205.69	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	97.83	76.77	205.69	0.00
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	118.97	27.37	229.51	1,084.38
Nutrient balance	-21.14	49.39	-23.82	-1,084.38
Applied to removed ratio	0.82	2.80	0.90	0.00

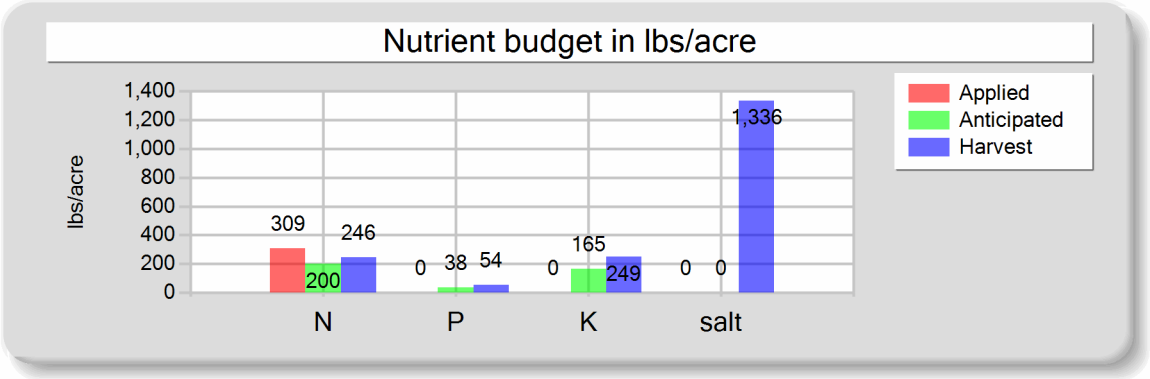
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6 - 06/05/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	200.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	101.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.68	0.00	0.00	0.00
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	245.56	53.63	249.17	1,336.14
Nutrient balance	63.12	-53.63	-249.17	-1,336.14
Applied to removed ratio	1.26	0.00	0.00	0.00

Fresh water applied	
21,888,000.00	*gallons*
806.06	*acre-inches*
26.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 08/04/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 12.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	13,100.00	35,100.00							
DL	100.00	100.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 01/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 62.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,900.00	7,050.00	12,200.00							
DL	100.00	10.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 07/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 15.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,700.00	8,020.00	11,800.00							
DL	100.00	10.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Denele Analytical

Sample and source description: Denele Analytical

Sample date: 01/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	98.00	30.80	0.00	0.00	40.00	885.00								3,780.00	3,260
DL	1.00	1.00	1.00	1.00	1.00	1.00								10.00	10

C. FRESH WATER ANALYSES

Ag 1

FGL Environmental

Sample description: FGL Environmental

Sample date: 04/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	11.30	0.00	11.30	74.00	39.00	40.00	350.00	0.00	41.80	58.00	826.00	540
DL	0.10	0.20	0.10	1.00	1.00	1.00	10.00	10.00	0.50	1.00	1.00	20

Ag 2

FGL Agricultural

Sample description: FGL Agricultural

Sample date: 04/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	16.70	0.00	16.70								1,000.00	
DL	0.10	0.20	0.10								1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/26/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,300.00	236.00	18,200.00		10.40
DL	100.00	10.00	100.00		0.10

1 - 06/05/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	2,660.00	20,800.00		9.60
DL	100.00	10.00	100.00		0.10

2 - 10/28/2009: Pasture

Estimate - Not sampled

Sample and source description: Estimate - Not sampled

Sample date: 09/30/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,000.00	2,000.00	10,000.00		
DL	1.00	1.00	1.00		

3 - 10/26/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,200.00	2,260.00	32,400.00		10.20
DL	100.00	10.00	100.00		0.10

3 - 06/05/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,650.00	19,400.00		8.10
DL	100.00	10.00	100.00		0.10

4 - 10/26/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	2,310.00	20,900.00		9.30
DL	100.00	10.00	100.00		0.10

4 - 06/05/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,500.00	2,210.00	19,200.00		8.20
DL	100.00	10.00	100.00		0.10

5 - 10/26/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,100.00	25,300.00		11.00
DL	100.00	10.00	100.00		0.10

5 - 06/05/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,180.00	14,600.00		7.80
DL	100.00	10.00	100.00		0.10

6 - 10/26/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,600.00	21,800.00		10.30
DL	100.00	10.00	100.00		0.10

6 - 06/05/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	2,970.00	13,800.00		7.40
DL	100.00	10.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

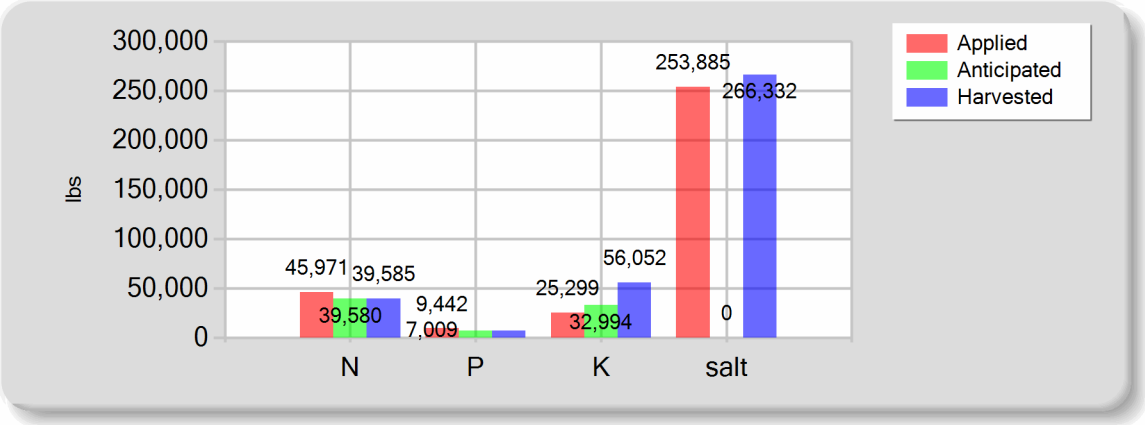
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

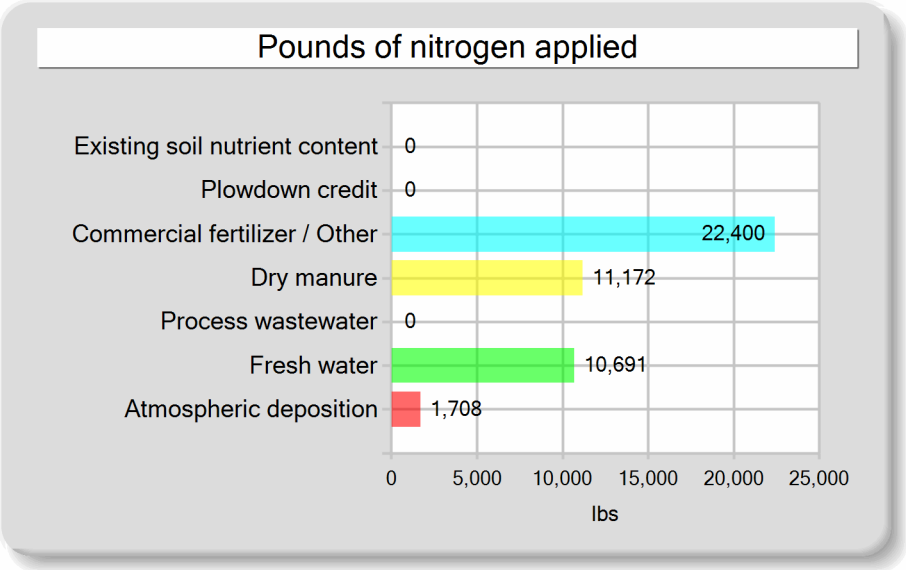
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

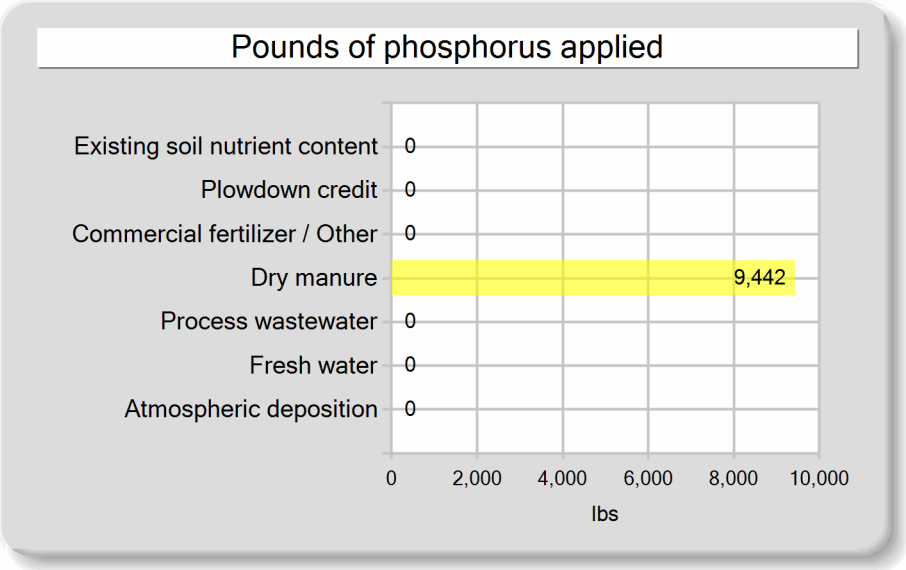
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	22,400.00	0.00	0.00	0.00
Dry manure	11,172.09	9,442.22	25,299.38	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10,691.02	0.00	0.00	253,884.94
Atmospheric deposition	1,708.00	0.00	0.00	0.00
Total nutrients applied	45,971.11	9,442.22	25,299.38	253,884.94
Anticipated crop nutrient removal	39,580.00	7,008.80	32,994.40	0.00
Actual crop nutrient removal	39,585.48	7,130.77	56,052.01	266,332.44
Nutrient balance	6,385.63	2,311.45	-30,752.63	-12,447.50
Applied to removed ratio	1.16	1.32	0.45	0.95

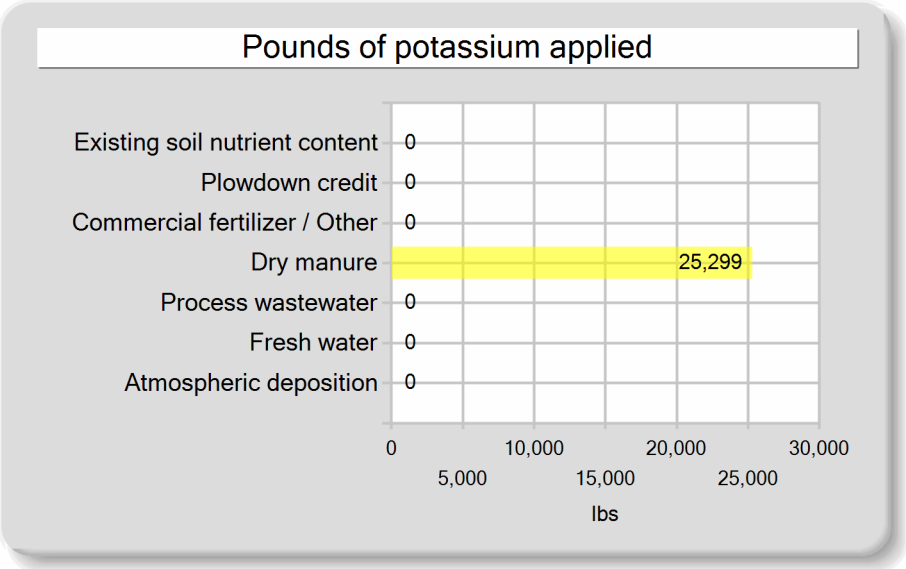
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

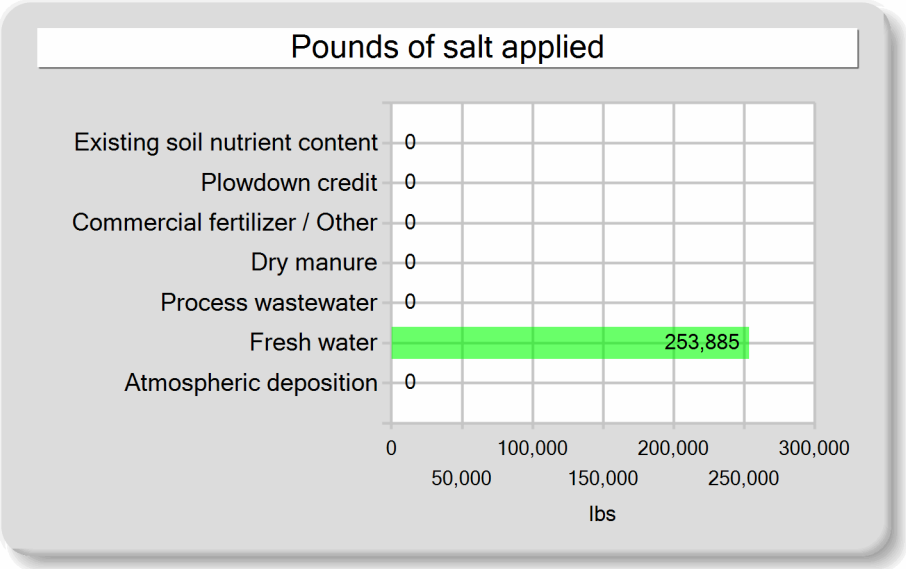


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Application area #2, pasture, was not sampled in 2023 due to operator error. No manure or manure water were applied to this field, nor was any commercial fertilizer applied.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
William Vander Meulen	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
William Vander Meulen	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-18-24	
DATE	DATE

ıy 15, 2023

Vander Meulen Dairy
Attn: Bill Vander Meulen
20250 E. Avena
Escalon, CA. 95320

Description : Ag 1
Project : Domestic & Ag Supply Wells

Lab No. : STK2334920-001
Customer No. : 3016366

Sampled On : April 20, 2023 at 11:30
Sampled By : Bill Vander Meulen
Received On : April 20, 2023 at 12:00
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Ammonia-N	ND	0.2	mg/L		1	U	05/04/2023	14:10	lcr	SM 4500-NH3 G	05/07/2023	19:23	lcr
Alkalinity (as CaCO3)	290	10	mg/L		1		05/03/2023	14:52	amm	SM 4500-H+B	05/04/2023	05:13	sta
Bicarbonate	350	10	mg/L		1		05/03/2023	14:52	amm	SM 4500-H+B	05/04/2023	05:13	sta
Carbonate	ND	10	mg/L		1	U	05/03/2023	14:52	amm	SM 4500-H+B	05/04/2023	05:13	sta
Hydroxide	ND	10	mg/L		1	U	05/03/2023	14:52	amm	SM 4500-H+B	05/04/2023	05:13	sta
Chloride	**58**	**1**	**mg/L**		**1**	**b**	**04/21/2023**	**15:31**	**ldm**	**EPA 300.0**	**04/22/2023**	**10:10**	**ldm**
Nitrate Nitrogen	11.3	0.1	mg/L		1		04/21/2023	15:31	ldm	EPA 300.0	04/22/2023	10:10	ldm
Conductivity	826	1	umhos/cm		1		05/03/2023	14:52	amm	SM 4500-H+B	05/04/2023	05:13	sta
Sulfate	31.8	0.5	mg/L		1		04/21/2023	15:31	ldm	EPA 300.0	04/22/2023	10:10	ldm
Solids, Total Dissolved (TDS)	540	20	mg/L		1		04/24/2023	14:00	ctl	SM 2540 C	04/25/2023	11:30	ctl
Calcium	**74**	**1**	**mg/L**		**1**		**04/27/2023**	**06:55**	**ejc**	**EPA 200.7**	**04/29/2023**	**12:50**	**ac**
Magnesium	39	1	mg/L		1		04/27/2023	06:55	ejc	EPA 200.7	04/29/2023	12:50	ac
Sodium	40	1	mg/L		1	h	04/27/2023	06:55	ejc	EPA 200.7	04/29/2023	12:50	ac

Flags Definition:
Constituent results were non-detect.
b The Blank was positive for constituent but less than the PQL
h The MS/MSD did not meet QC criteria.

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution

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9415 W. Goshen Avenue
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CA ELAP Certification No. 2

ıy 15, 2023

Lab No. : STK2334920-002
Customer No. : 3016366

Vander Meulen Dairy
Attn: Bill Vander Meulen
20250 E. Avena
Escalon, CA. 95320

Sampled On : April 20, 2023 at 11:30
Sampled By : Bill Vander Meulen
Received On : April 20, 2023 at 12:00
Matrix : Ag Water

Description : Ag 2
Project : Domestic & Ag Supply Wells

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
Dairy Analysis							Date	Time	Who	Method	Date	Time	Who
Ammonia-N	ND	0.2	mg/L		1	U	05/04/2023	14:10	lcr	SM 4500-NH3 G	05/07/2023	19:25	lcr
Nitrate Nitrogen	16.7	0.1	mg/L		1		04/21/2023	15:31	ldm	EPA 300.0	04/22/2023	05:59	ldm
Conductivity	**1000**	**1**	**umhos/cm**		**1**		**05/09/2023**	**18:02**	**amm**	**SM 4500-H+B**	**05/09/2023**	**22:07**	**amm**

DQF Flags Definition:
U Constituent results were non-detect.

ND=Non-Detected, RL=Reporting Level , Dil.=Dilution

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ıy 15, 2023

Vander Meulen Dairy
Attn: Bill Vander Meulen
20250 E. Avena
Escalon, CA. 95320

Description : Domestic 1
Project : Domestic & Ag Supply Wells

Lab No. : STK2334920-003
Customer No. : 3016366

Sampled On : April 20, 2023 at 11:30
Sampled By : Bill Vander Meulen
Received On : April 20, 2023 at 12:00
Matrix : Ag Water

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
							Date	Time	Who	Method	Date	Time	Who
Dairy Analysis													
Ammonia-N	ND	0.2	mg/L		1	U	05/04/2023	14:10	lcr	SM 4500-NH3 G	05/07/2023	19:27	lcr
Nitrate Nitrogen	8.0	0.3*	mg/L		3		04/21/2023	15:31	ldm	EPA 300.0	04/22/2023	11:13	ldm
Conductivity	689	1	umhos/cm		1		05/09/2023	18:02	amm	SM 4500-H+B	05/09/2023	22:01	amm

DQF Flags Definition:
U Constituent results were non-detect.

ND=Non-Detected, RL=Reporting Level * RL adusted for dilution, Dil.=Dilution

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ay 15, 2023

Lab No. : STK2334920-004
Customer No. : 3016366

Vander Meulen Dairy
Attn: Bill Vander Meulen
20250 E. Avena
Escalon, CA. 95320

Sampled On : April 20, 2023 at 11:30
Sampled By : Bill Vander Meulen
Received On : April 20, 2023 at 12:00
Matrix : Ag Water

Description : Domestic 2
Project : Domestic & Ag Supply Wells

Sample Results - Inorganic

Constituent	Result	RL	Units	Note	Dil.	DQF	Sample Preparation			Sample Analysis			
							Date	Time	Who	Method	Date	Time	Who
Dairy Analysis													
Ammonia-N	**ND**	**0.2**	**mg/L**		**1**	**U**	**05/04/2023**	**14:10**	**lcr**	**SM 4500-NH3 G**	**05/07/2023**	**19:30**	**lcr**
Nitrate Nitrogen	16.6	0.1	mg/L		1		04/21/2023	15:31	ldm	EPA 300.0	04/22/2023	04:56	ldm
Conductivity	1000	1	umhos/cm		1		05/09/2023	18:02	amm	SM 4500-H+B	05/09/2023	21:58	amm

DQF Flags Definition:
U Constituent results were non-detect.

ND=Non-Detected, RL=Reporting Level * RL adusted for dilution, Dil.=Dilution

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FGL ENVIRONMENTAL & AGRONOMIC

Dairy Operations
www.fglinc.com

CHAIN OF CUSTODY AND ANALYSES REQUEST FORM

CLIENT DETAILS — SECTION I

Client: VanDer Meulen Dairy

☐ New Customer Customer Number: 3016366

Address: Attn: Bill Vander Meulen
20250 E. Avena
Escalon, CA. 95320

Phone: (209)838-2606 FAX:
Email:
Project Name: Domestic & Ag Supply Wells
Contact person: Bill Vander Meulen

Billing Information (if different from above)

Name:
Address:
Phone: FAX:
Contact Person:
Purchase order/contract/FGL quote number:

SAMPLING — SECTION II

Lab Number: STK2334920
Time:
Shipping Charge:

Sampler(s): Bill VanDer Meulen
Mileage:
Pickup Charge:

SAMPLE INFORMATION — SECTION III

Samp Num	Location/Description	Date Sampled	Time Sampled	Number of Containers	Type of Container:(G)Glass, (P) Plastic, (Z) Ziplock, (PB) Paper Bag	(PW) Process Water, (M)Manure,(S) Soil, (T) Crop Tissue, (IW)Irrigation Water, (SW) Storm Water	(1) H2S04, (2) HNO3, (3) Na2S203, (4) Other	DOMESTIC & AG SUPPLY WELLS	DAW1: NO3-N; NH3-N; EC	DAW1 (if detected in field): NH3-N	DAW2: Alk. (CaCO3),Cl,TDS,SO4,Ca,Na,Mg	DAW Field-Field E.C.	DAW Field-Field NH3-N						
1	Ag 1	4-20-23	1130						X		X								
2	Ag 2																		
3	Domestic 1																		
4	Domestic 2																		

ANALYSES REQUESTED

REMARKS — SECTION IV

CUSTODY — SECTION V

	Print Name	Date	Time
Reliquished By:		4-20-23	11:30
Received By:	Tracy Guadalupe	4-20-23	1200
Reliquished By:		4/20/23	1700
Received By:		4/20/23	1700

GLS CDA 4/21/23 1311

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Field Office
Visalia, California
Cell: (559) 737-2399
Tel: (559) 734-9473
FAX: (559) 734-8435

Inter-Laboratory Condition Upon Receipt (Attach to COC) 2334920

Sample Receipt at: STK CC CH VI

1. Number of ice chests/packages received: 1 Shipping tracking # _____
2. Were samples received in a chilled condition? Temps: 14.8 / ____ / ____ / ____ / ____
   Surface water SWTR bact samples: A sample that has a temperature upon receipt of >10° C, whether iced or not, should be flagged unless the time since sample collection has been less than two hours.
3. Do the number of bottles received agree with the COC? Yes No N/A
4. Were samples received intact? (i.e. no broken bottles, leaks etc.) Yes No
5. VOAs checked for Headspace? Yes No N/A
6. Were sample custody seals intact? Yes No N/A
7. If required, was sample split for pH analysis? Yes No N/A
8. Were all analyses within holding times at time of receipt? Yes No
9. Verify sample date, time and sampler name Yes No

Sign and date the COC, place in a ziplock and put in the same ice chest as the samples.
Sample Receipt Review completed by (initials): TAC

Sample Receipt at SP:

1. Were samples received in a chilled condition? Temps: 2 / 2 / 3 / 3 / ____
   Acceptable is above freezing to 6⊘ C. If many packages are received at one time check for tests/H.T.'s/rushes/
2. Shipping tracking numbers:
   559243595 / 307 / 845 / 4461
3. Do the number of bottles received agree with the COC? Yes No N/A
4. Were samples received intact? (i.e. no broken bottles, leaks etc.) Yes No
5. Were sample custody seals intact? Yes No N/A

Sign and date the COC, obtain LIMS sample numbers, select methods/tests and print labels.

Sample Verification, Labeling and Distribution:

1. Were all requested analyses understood and acceptable? Yes No
2. Did bottle labels correspond with the client's ID's? Yes No
3. Were all bottles requiring sample preservation properly preserved? Yes No N/A FGL
   [Exception: Oil & Grease, VOA and CrVI verified in lab]
4. VOAs checked for Headspace? Yes No N/A
5. Have rush or project due dates been checked and accepted? Yes No N/A
6. Were all analyses within holding times at time of receipt? Yes No

Attach labels to the containers and include a copy of the COC for lab delivery.
Sample Receipt, Login and Verification completed by (initials): CDA

Discrepancy Documentation:

Any items above which are "No" or do not meet specifications (i.e. temps) must be resolved.

1. Person Contacted: _____ Phone Number: _____
   Initiated By: _____ Date: _____
   Problem:
   Resolution:

2. Person Contacted: _____ Phone Number: _____
   Initiated By: _____
   Problem:
   Resolution:

(Please use the back of this sheet for additional con
contacts)

(3016366)
VanDer Meulen Dairy
STK2334920
cda 04/21/2023 10:43:34